

[Academic Presentation Title]

[Presentation Subtitle]

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[Conference Name, Month Year]

Progressive vs. Non-progressive Visualization

Non-progressive visualization

- Renders in one pass.
- Each change in the data, whether caused by interaction or new data, triggers a full rendering.

Progressive visualization

- Rendering uses a series of passes that improve the result until completion.
- Each pass is limited to a requested latency and renders a meaningful partial result.
- Each partial result has a meaningful error or quality measure and an estimate of completion, such as a progress bar.
- Users can interact at any time; the visualization adapts as much as possible and can implement steering when needed.

Definition adapted from Ulmer et al., “A Survey on Progressive Visualization” (2024).

Progressive Computation

F_p : progressive function

q : requested interval between outputs

R_{i-1}, R_i, R : previous, partial, and final result

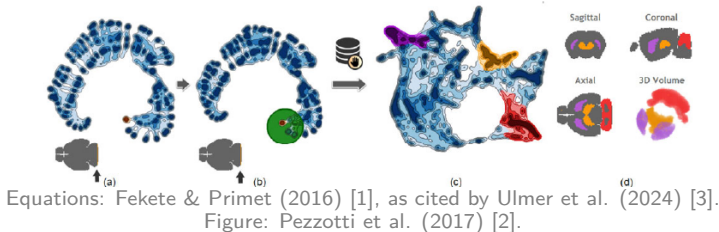
P : function parameters; D_i : data at pass i

t_i, m_i : time and memory at pass i

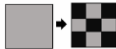
$R_0 = \emptyset$: first call has no previous result

$$F_p(P, D_i, q, R_{i-1}) \xrightarrow{t_i, m_i} R_i, \quad i = 1, \dots, z$$

$t_i \leq q$, $R_i \rightarrow R$ or a termination criterion is reached.



Three Ways to Produce Progressive Results

Visualization Categories	Temporal 	Geo-Spatial 	Hierarchical 	Network 	Multidimensional 	Field 
Progressive Processing Categories	Data Chunking 		Process Chunking 		Custom Chunking 	
Data Domain Categories	Known End 			Unknown End 		
Visual Update Pattern	Extension 			Overwrite 		

Ulmer et al. (2024), "A Survey on Progressive Visualization."

Data chunking

Add data chunks successively. R_i reflects the data processed so far.

Process chunking

Iterate over loaded data. R_i is the algorithm's current output.

Custom chunking

Use a tailored or hybrid schedule beyond the two basic modes.

A Taxonomy of Progressive Visualizations

- **Visualization type:** temporal, geo-spatial, hierarchical, network, multidimensional, or field.
- **Processing:** data chunking, process chunking, or custom chunking.
- **Data domain:** known end or unknown end.
- **Visual update:** extend the current view or overwrite it with a new state.

Source: Ulmer et al. (2024).

Progressive Dimensionality Reduction

- t-SNE builds an embedding through iterative optimization, but it must first compute high-dimensional similarities.
- Barnes-Hut t-SNE reduces optimization cost, yet initialization can still delay the first interpretable embedding.
- Q-SNE approximated neighbors for sparse document data; A-tSNE extends approximate similarities to dense high-dimensional data.

Source: Pezzotti et al. (2017), Related Work.

Field Trends and Open Challenges

- The survey classifies 48 papers from 1999 to 2023; 33 of 48 use multidimensional visualization.
- Steering appears in 35 studies, uncertainty in 20, and real-time processing in 10.
- Open challenges include visual stability as estimates change, uncertainty communication, and evaluation beyond runtime.

Source: Ulmer et al. (2024), study set and discussion.

Synthesis and Paper Positioning

- A-tSNE addresses an algorithm-level bottleneck: delayed initialization in high-dimensional t-SNE.
- The 2024 survey offers a field-level taxonomy and reviews evaluation methods and research gaps.
- **Synthesis:** progressive methods must make partial results timely, interpretable, steerable, and stable as they improve.

Sources: Pezzotti et al. (2017); Ulmer et al. (2024).

Paper 1: Overview

- *[Title, authors, and publication details]*
- *[Research subject and context]*
- *[Core question addressed in this paper]*

Paper 1: Motivation

- *[Why the research question matters]*
- *[Limitations of existing explanations]*
- *[How the paper approaches the question]*

Paper 1: Research Questions and Hypotheses

- *[Research question]*
- *[Hypothesis or proposition 1]*
- *[Hypothesis or proposition 2, if applicable]*

Paper 1: Conceptual Framework

- *[Key concepts or variables]*
- *[Theoretical relationships and proposed mechanisms]*
- *[Insert the paper's central framework figure here]*

Paper 1: Data and Research Subjects

- *[Data or material sources and time period]*
- *[Sample composition or case selection]*
- *[Data processing and ethics, if applicable]*

Paper 1: Research Methods

- *[Research design and analytical method]*
- *[Model, experiment, or analysis workflow]*
- *[Key identification assumptions or derivation conditions]*

Paper 1: Results Overview

- *[Main result 1]*
- *[Main result 2]*
- *[How the results address the hypotheses or propositions]*

Paper 1: Main Evidence

- *[Insert the key figure, table, or theorem]*
- *[Key estimate, proof, or observation]*
- *[Conclusion supported by this evidence]*

Paper 1: Mechanisms and Additional Analyses

- *[Mechanism analysis or theoretical explanation]*
- *[Robustness, sensitivity, or alternative explanation checks]*
- *[Boundary conditions for the results]*

Paper 1: Summary

- *[Main conclusion]*
- *[Contribution and limitation]*
- *[Connection to Paper 2]*

Paper 2: Overview

- *[Title, authors, and publication details]*
- *[Research subject and context]*
- *[Core question addressed in this paper]*

Paper 2: Research Question and Contribution

- *[Research question]*
- *[What this paper adds to related work]*
- *[Relationship to or distinction from Paper 1]*

Paper 2: Framework and Hypotheses

- *[Theoretical basis or conceptual framework]*
- *[Hypotheses or propositions]*
- *[Proposed mechanism or reasoning]*

Paper 2: Data and Research Design

- *[Data or material sources and sample]*
- *[Research design and key variables or concepts]*
- *[Why the design fits the research question]*

Paper 2: Methods and Analysis Strategy

- *[Model, experiment, comparison, or proof method]*
- *[Identification strategy or analytical steps]*
- *[Key assumptions and how they are assessed]*

Paper 2: Main Results

- *[Main result 1 and supporting evidence]*
- *[Main result 2 and supporting evidence]*
- *[How the results answer the research question]*

Paper 2: Mechanisms and Robustness

- *[Mechanisms, heterogeneity, or boundary conditions]*
- *[Robustness or sensitivity analyses]*
- *[Alternative explanations and additional evidence]*

Paper 2: Summary and Overall Conclusions

- *[Main conclusion of Paper 2]*
- *[Shared findings and differences across the two papers]*
- *[Overall answer to the research topic]*

- [1] J.-D. Fekete and R. Primet, “Progressive Analytics: A Computation Paradigm for Exploratory Data Analysis,” CoRR, abs/1607.05162, 2016.
- [2] N. Pezzotti et al., Approximated and User Steerable t-SNE for Progressive Visual Analytics, *IEEE Transactions on Visualization and Computer Graphics*, 23(7), 1739–1752, 2017. doi:10.1109/TVCG.2016.2570755.
- [3] A. Ulmer et al., A Survey on Progressive Visualization, *IEEE Transactions on Visualization and Computer Graphics*, 30(9), 6447–6467, 2024. doi:10.1109/TVCG.2023.3346641.

[Thank You]

[Questions and Discussion]

[Presenter email, optional]